PID/DATA MONITOR TABLE [START STOP UNIT]

id0902p6025500

—: Not applicable

PID	Unit/ Operation	Data contents	Data read/use method	Module control terminal
BRAKE_SW1*1	Off/On	Brake switch (No.1 signal) signal input status • Off: Brake switch (No.1 signal) is off. • On: Brake switch (No.1 signal) is on.	—	1C (Brake switch)
CLUT_CUT_SW*2	Off/On	Starter interlock switch input status • Off: Starter interlock switch is off. • On: Starter interlock switch is on.	—	1F (Starter interlock switch)
CLUTCH_SW*2	Off/On	Clutch switch input status • Off: CPP switch is off. • On: CPP switch is on.	—	2E (Clutch pedal position switch)
COMM_ST_TSP	Off/On	Displays operation condition of the immobilizer system key code read function • Off: Key code read function not performed • On: Code read function is performed and coil antenna reads key code from remote transmitter transponder.	Key code reading function of immobilizer system is performed and when a brake switch, clutch switch, or push button start signal is input, the coil antenna reads the key code from the transponder of the remote transmitter, and displays On while the key code is being sent to the start stop unit.	• 1AC (PATS+) • 1AE (PATS-)
CRU_CON_SW*3	Not_Pressed/On/Off/ On/Off/ Cancel/ Resume/ Fault/Set (+)/Set (-)/ Distance (+)/ Distance (-)	Cruise control switch operation condition • Not_Pressed: Each switch of cruise control switch is not pressed or speed limiter switch is pressed. • On: Cruise ON switch is pressed. • Off: Cruise OFF switch is pressed. • On/Off: — • Cancel: CANCEL switch is pressed. • Resume: RESUME (RES) switch is pressed. • Fault: Cruise control switch has a malfunction. • Set (+): SET (+) switch is pressed. • Set (-): SET (-) switch is pressed. • Distance (+): — • Distance (-): —	—	1Y (Cruise control switch signal)
ES_UL_SW	Not_Unlock/ Unlock	Steering lock unlock switch condition • Not_Unlock: Steering lock unlock switch is off. • Unlock: Steering lock unlock switch is on.	—	—
ES_UL_SW_M	Not_Unlock/ Unlock	Memory value of steering lock unlock switch • Not_Unlock: Memory value of steering lock unlock switch is not unlocked. • Unlock: Memory value of steering lock unlock switch is unlocked.	—	—
ES_ULL_SW	Off/On	Steering lock unlock/lock switch condition • Off: Steering lock unlock/lock switch is off. • On: Steering lock unlock/lock switch is on.	—	—

PID	Unit/ Operation	Data contents	Data read/use method	Module control terminal
ES_ULL_SW_M	Off/On	Memory value of steering lock unlock/lock switch • Off: Memory value of steering lock unlock/lock switch is off. • On: Memory value of steering lock unlock/lock switch is on.	—	—
F_FOG_SW	Off/On	Front fog light switch operation condition • Off: Fog light switch is not in F.FOG position. • On: Fog light switch is in F.FOG position.	—	—
H/L_SW_HI	Off/On	Headlight switch (HI) operation condition • Off: Light switch is not in HI position. • On: Light switch is in HI position.	—	—
H/L_SW_LOW	Off/On	Displays dimmer switch position of the light switch that is being input to the start stop unit • Off: Dimmer switch of light switch is not in LO position • On: Dimmer switch of light switch is in LO position	Verify if the monitor value switches when the dimmer switch of the light switch is switched between LO and HI positions. If the dimmer switch operation and monitor values do not match, inspect the light switch.	—
H/L_SW_OFF	Off/On	Headlight light switch (OFF) operation condition • Off: Light switch is not in OFF position. • On: Light switch is in OFF position.	—	—
H/L_SW_PASS	Off/On	Headlight switch (PASS) operation condition • Off: Light switch is not in passing position. • On: Light switch is in passing position.	—	—
H/L_SW_TNS	Off/On	Headlight switch (TNS) operation condition • Off: Light switch is not in TNS position. • On: Light switch is in TNS position.	—	—
HAZARD_SW	Off/On	Displays condition of hazard warning switch input to start stop unit • Off: Hazard warning switch is off. • On: Hazard warning switch is on.	—	1G (Hazard warning switch signal)
IG_POSITION	OFF/ACC/ ON/CRANK	Displays power status controlled by the start stop unit • OFF: Start stop unit controls ignition switch off • ACC: Start stop unit controls ignition switch ACC • ON: Start stop unit controls ignition switch ON (engine off) • CRANK: Start stop unit controls cranking at ignition switch ON	• When the push button start is operated, the start stop unit performs power source switching and displays the PID value of the power source position in which control was performed. • If the push button start operation and PID value do not match, and a power source status of another unit differs from the PID value, inspect the input power source of the other unit because the power source control of the start stop unit is normal.	—

PID	Unit/ Operation	Data contents	Data read/use method	Module control terminal
INFO_SW*4	Off/ SW1_On/ SW2_On/ SW3_On/ Unknown/ Invalid	Steering switch operation condition • Off: Steering switch UP/DOWN/INFO switch is not pressed. • SW1_On: UP switch is pressed. • SW2_On: DOWN switch is pressed. • SW3_On: INFO switch is pressed. • Unknown: Steering switch UP/DOWN/ INFO switch signal is not determined. • Invalid: Steering switch malfunction is received.	Displays INFO switch position of steering switch input to start stop unit. Steering switch signal is input to start stop unit via clock spring.	• 1AA (Steering switch signal) • 1AB (Steering switch ground)
INHIBIT_SW*1	Off/On	Transaxle range switch input status • Off: Selector lever is not in P and N positions. • On: Selector lever is in P or N position.	—	1F (Selector lever position switch)
LG/T_LK_SW*5	Off/On	Request switch (liftgate) operation condition • Off: Request switch (liftgate) is off. • On: Request switch (liftgate) is on.	—	2F (LF control unit communication)
LG/T_UNL_SW	Off/On	**4SD** • Trunk lid opener switch operation condition — Off: Trunk lid opener switch is off. — On: Trunk lid opener switch is on. **5HB** • Liftgate opener switch operation condition — Off: Liftgate opener switch is off. — On: Liftgate opener switch is on.	—	1Q (BCM communication)
LL_SW_D_LK	Off/On	Driver's door lock link switch (lock side) operation condition • Off: Front door lock-link switch (driver's side) is in unlock. • On: Front door lock-link switch (driver's side) is in lock.	—	1P (Lock input)
NUM_TRNSMIT	—	Displays number of the remote transmitters programmed to the start stop unit	—	—
P_RANGE_SW*1	Off/On	Not P position switch condition • Off: NOT P position switch is off. • On: NOT P position switch is on.	—	1J (Not P position switch)
PAD_DOWN_SW* 6	Off/On	Steering shift switch (DOWN) condition • Off: DOWN switch of steering shift switch is off. • On: DOWN switch of steering shift switch is on.	—	1N (Steering shift switch (down))
PAD_UP_SW*6	Off/On	Steering shift switch (UP) condition • Off: UP switch of steering shift switch is off. • On: UP switch of steering shift switch is on.	—	1L (Steering shift switch (up))
R_FOG_SW	Off/On	Rear fog light switch condition • Off: Fog light switch is not in R.FOG position. • On: Fog light switch is in R.FOG position.	—	—
RELAY_ACC	Off/On	ACC relay operation condition • Off: ACC relay drive is off. • On: ACC relay drive is on.	—	2C (ACC relay control)
RELAY_IG1	Off/On	IG1 relay operation condition • Off: IG1 relay drive is off. • On: IG1 relay drive is on.	—	2S (IG1 relay control)
RELAY_IG2	Off/On	IG2 relay operation condition • Off: IG2 relay drive is off. • On: IG2 relay drive is on.	—	2D (IG2 relay control)

PID	Unit/ Operation	Data contents	Data read/use method	Module control terminal
RF_RECEP_ST	Initial/ Transmitter/ RF_Response	Displays radio reception status from the remote transmitter • Initial: Signal from remote transmitter is not received by start stop unit • Transmitter: Signal by button operation on remote transmitter is received by start stop unit • RF_Response: During remote transmitter authorization, response signal from remote transmitter is received by start stop unit	When verifying the PID value while operating the remote transmitter buttons, if “Transmitter” is displayed it can be determined that the transmission between the remote transmitter, keyless receiver, and start stop unit is normal.	2R (Keyless receiver communication)
RF_TRNS_NUM	Initial/No.1/ No.2/No.3/ No.4/No.5/ No.6	When pressing the unlock button of the remote transmitter, the registration number of the remote transmitter sending radio waves to the start stop unit is displayed. • Initial: Registration number of remote transmitter cannot be received • No.1: Registration number No.1 remote transmitter is received • No.2: Registration number No.2 remote transmitter is received • No.3: Registration number No.3 remote transmitter is received • No.4: Registration number No.4 remote transmitter is received • No.5: Registration number No.5 remote transmitter is received • No.6: Registration number No.6 remote transmitter is received	When DTC B13D3:16 (Low remote transmitter battery voltage) is detected, the registration number of the remote transmitter in which the battery voltage has decreased is displayed in the snapshot data item TRNS_NUM_L. Display the registration number of the remote transmitter using the PID, verify the remote transmitter with the low battery voltage, and inspect the battery condition of the target remote transmitter.	2R (Keyless receiver communication)
RQ_SW_LF*5	Off/On	Request switch (LF) operation condition • Off: Request switch (LF) is off. • On: Request switch (LF) is on.	The signal is input to the start stop unit through the LF control unit of the request switch (LF).	2F (LF control unit communication)
RQ_SW_RF*5	Off/On	Request switch (RF) operation condition • Off: Request switch (RF) is off. • On: Request switch (RF) is on.	The signal is input to the start stop unit through the LF control unit of the request switch (RF).	2F (LF control unit communication)
SECURITY_I	Off/On	Output condition of security indicator • Off: Security indicator light is turned off. • On: Security indicator light is illuminated.	Security indicator on request to instrument cluster	2M, 2O (CAN)
SFT_LK_SOL*1	Off (Lock)/ On (Unlock)	Shift lock solenoid operation condition • Off (Lock): Shift-lock is operated. • On (Unlock): Shift-lock is released.	—	2G (Shift lock)
SPD_LIM_SW*7	Off/On/ Unknown	Speed limiter switch input status • Off: Speed limiter switch is off. • On: Speed limiter switch is on. • Unknown: Speed limiter switch signal is not determined.	—	1Y (Cruise control switch signal)
SSB_1	Off/On	Push button start switch 1 input signal • Off: Push button start switch 1 is off. • On: Push button start switch 1 is on.	—	1H (Push button start 1)
SSB_2	Off/On	Push button start switch 2 input signal • Off: Push button start switch 2 is off. • On: Push button start switch 2 is on.	—	2J (Push button start 2)
SSB_AMBER	Off/On	Push button start indicator light (amber) illumination request • Off: Push button start indicator light (amber) is turned off. • On: Push button start indicator light (amber) is illuminated.	—	1M (Push button illumination (amber))

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SSB_GREEN	Off/On	Push button start indicator light (green) illumination request • Off: Push button start indicator light (green) is turned off. • On: Push button start indicator light (green) is illuminated.	—	1K (Push button illumination (green))
SSB_ILLMI	Off/On	Push button start illumination request • Off: START/STOP/ENGINE indicator light is turned off. • On: START/STOP/ENGINE indicator light is illuminated.	—	1I (Push button illumination)
START_RLY_1	V	Displays voltage at start stop unit terminal 2V	—	2V (Starter relay)
START_RLY_2	V	Displays voltage at start stop unit terminal 1D	—	1D (Starter monitor)
STR_A_ANG_A*8	° (deg)	Displays steering angle sensor A signal aiming angle	—	—
STR_A_ANG_B*8	° (deg)	Displays steering angle sensor B signal aiming angle	—	—
STR_AB_ANG*8	° (deg)	Displays steering angle signal (absolute angle) • Steering wheel in neutral position: Near **0 degrees** • Steering wheel turned to left: Changes from **0 degrees** to positive • Steering wheel turned to right: Changes from **0 degrees** to negative	—	—
STR_ANG_S*8	Without/ With	Vehicle with/without steering angle sensor • Without: Steering angle sensor is not equipped. • With: Steering angle sensor is equipped.	—	—
STR_ANG_S_A*8	V	Displays voltage at the steering angle sensor A signal	—	1U (Steering angle sensor (signal A/ D1)) **Note** • (See STEERING ANGLE SENSOR INSPECTION.)
STR_ANG_S_B*8	V	Displays voltage at the steering angle sensor B signal	—	1W (Steering angle sensor (signal A/ D1)) **Note** • (See STEERING ANGLE SENSOR INSPECTION.)
STR_ANG_S_V*8	V	Displays steering angle sensor power supply voltage **Note** • “5 V” is displayed even if the voltage is **5 V or more**.	—	1S (Steering angle sensor power supply)
T_SEN_LF	Off/On	Displays in the M-MDS but it does not operate.		
T_SEN_RF	Off/On	Displays in the M-MDS but it does not operate.		

PID	Unit/ Operation	Data contents	Data read/use method	Module control terminal
TURN_SW_L	Off/On	Turn switch (LH) input • Off: Turn switch is in position other than LH. • On: Turn switch is in LH position.	—	—
TURN_SW_R	Off/On	Turn switch (RH) input • Off: Turn switch is in position other than RH. • On: Turn switch is in RH position.	—	—
VPWR_ACC	V	Displays ACC power supply voltage input to start stop unit	If there is a malfunction in the ACC power supply voltage input to the start stop unit, the keyless entry system and push button start system do not operate normally because PID item VPWR_ACC is used for ACC power on/off determination.	1E (ACC monitor)
VPWR_B1	V	Displays B+ power supply (ROOM fuse) voltage input to start stop unit	If there is a malfunction in the B+ power supply (ROOM fuse) voltage input to the start stop unit, the keyless entry system does not operate normally because PID item VPWR_B1 is used for determining whether or not there is a ROOM fuse inserted.	1A (Power supply)
VPWR_B2	V	Displays B+ power supply (AT fuse) voltage input to start stop unit	When the ROOM fuse is removed, the B+ power supply (AT fuse) is the start stop unit power supply. If there is a malfunction in the B+ power supply (AT fuse) voltage input to the start stop unit when the ROOM fuse is removed, the push button start system does not operate normally.	2W (Power supply)
VPWR_B3	V	Displays B+ power supply (SRS2/ESCL fuse) input to start stop unit.	If there is a malfunction in the B+ power supply (SRS2/ESCL fuse) voltage input to the start stop unit, the steering lock and the push button start system do not operate normally because PID item VPWR_B3 is used as the steering lock power supply built into the start stop unit.	2X (Power supply)

PID	Unit/ Operation	Data contents	Data read/use method	Module control terminal
VPWR_IG1	V	Displays IG1 power supply voltage input to start stop unit	If there is a malfunction in the IG1 power supply voltage input to the start stop unit, the keyless entry system and the push button start system do not operate normally because PID item VPWR_IG1 is used for IG1 power on/off determination.	1B (Power supply)
WASHER_F	Off/On	Windshield washer switch signal input status • Off: Windshield washer switch is in OFF position. • On: Windshield washer switch is in ON position.	—	—
WASHER_R	Off/On	Rear washer switch signal input status • Off: Rear washer switch is not in ON position. • On: Rear washer switch is in ON position.	—	—
WIP_F_INT	Off/On	Windshield wiper switch (INT) signal input status • Off: Windshield wiper switch is in MIST, OFF or LO position. • On: Windshield wiper switch is in AUTO*9/INT*10 or HI position.	—	—
WIP_F_LOW	Off/On	Windshield wiper switch (LO) signal input status • Off: Windshield wiper switch is in OFF or AUTO*9/INT*10 position. • On: Windshield wiper switch is in MIST, LO or HI position.	—	—
WIP_INT_E_A	Low/High	Wiper INT encoder A signal input status • Low: Windshield wiper switch INT encoder A value is low. • High: Windshield wiper switch INT encoder A value is high.	—	—
WIP_INT_E_B	Low/High	Wiper INT encoder B signal input status • Low: Windshield wiper switch INT encoder B value is low. • High: Windshield wiper switch INT encoder B value is high.	—	—
WIP_INT_E_C	Low/High	Wiper INT encoder C signal input status • Low: Windshield wiper switch INT encoder C value is low. • High: Windshield wiper switch INT encoder C value is high.	—	—
WIP_R_INT	Off/On	Rear wiper switch (INT) signal input status • Off: Rear wiper switch is not in INT position. • On: Rear wiper switch is in INT position.	—	—
WIP_R_ON	Off/On	Rear wiper switch (ON) signal input status • Off: Rear wiper switch is not in ON position. • On: Rear wiper switch is in ON position.	—	—

*1 : ATX
*2 : MTX
*3 : With cruise control system

*4 : With steering switch
*5 : With advanced keyless entry system
*6 : With steering shift switch
*7 : With adjustable speed limiter (ASL)
*8 : With smart city brake support (SCBS)
*9 : With auto wiper system
*10 : Without auto wiper system